

Name

- Show appropriate work.
  - Appropriate work for a linear ODE includes the integrating factor and some intermediate steps.
  - Appropriate work for a missing eigenvector includes the linear system you are solving and the solution.
  - Labelling and using eigenvalues/eigenvectors etc. appropriately is a good idea.
- Explicitly real solutions are required.
- Many questions ask only for a linear system. Do not solve the system in these cases.
- Eigenvalues and eigenvectors are given. Do not attempt to recompute them.

## Linear ODEs

### Section 2.3

1. Solve the ODE  $2y'(t) - 4y(t) = -6e^{2t}$ .
2. Solve the ODE  $2y'(t) - 4y(t) = -6e^{2t}$ .
3. Solve the ODE  $2y'(t) - 4y(t) = -6e^{2t}$  with  $y(0) = 2$
4. Solve the ODE  $3y'(t) - 9y(t) = 6e^{3t}$ .
5. Solve the ODE  $3y'(t) + 6y(t) = -3e^{2t}$  with  $y(0) = -3$
6. Solve the ODE  $ty'(t) - 3y(t) = \frac{2}{t^3}$
7. Solve the ODE  $ty'(t) - 3y(t) = -3t^2$ .
- ✓ 8. Solve the ODE  $ty'(t) + 2y(t) = 2t^2$  with  $y(1) = 2$  (at end)
- ✓ 9. Solve the ODE  $ty'(t) + 2y(t) = 2t^2$ .
- ✓ 10. Solve the ODE  $ty'(t) + 3y(t) = 3t^3$  with  $y(1) = -1$  (ct end)

All with a box saying

$y(t) =$

## Complementary Solutions from Eigenvalues/Eigenvectors

## Sections 3.3, 3.4, & 3.5

Any and all of the questions from these sections can include. "Discuss the behavior as  $t \rightarrow \infty$ ."

1. Compute the complementary solution of  $y' = Ay$  where A is below

MatrixForm[Eigensystem[A =  $\begin{pmatrix} -5 & 6 & -9 \\ -3 & 4 & -9 \\ 0 & 0 & -2 \end{pmatrix}$ ]]

Out[8]//MatrixForm=

$\begin{pmatrix} -2 & -2 & 1 \\ \{-3, 0, 1\} & \{2, 1, 0\} & \{1, 1, 0\} \end{pmatrix}$

$\lambda_3$   
 $v_3$   
 $v_2$

$$y_c(t) = c_1 e^{-2t} \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} + c_2 e^{\lambda_2 t} v_2 + c_3 e^{\lambda_3 t} v_3$$

$$v_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$e^{\lambda_3 t} v_3$  matters most.  
top 2 components go to 0  
like  $e^t$ .

2. Compute the complementary solution of  $y' = Ay$  where A is below

MatrixForm[Eigensystem[A =  $\begin{pmatrix} 3 & 0 & -5 \\ 5 & -2 & -5 \\ 0 & 0 & -2 \end{pmatrix}$ ]]

Out[82]//MatrixForm=

$\begin{pmatrix} 3 & -2 & -2 \\ \{1, 1, 0\} & \{1, 0, 1\} & \{0, 1, 0\} \end{pmatrix}$

$$y_c(t) = \sum_{i=1}^3 c_i e^{\lambda_i t} v_i$$

$$\begin{aligned} \lambda_1 &= 3 \\ \lambda_2 &= -2 \\ \lambda_3 &= -2 \end{aligned}$$

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\sim c_1 e^{3t} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ as } t \rightarrow \infty$$

3. Compute the complementary solution of  $y' = Ay$  where A is below

MatrixForm[Eigensystem[A =  $\begin{pmatrix} 4 & 7 & -12 \\ 4 & 8 & -8 \\ 2 & 5 & -6 \end{pmatrix}$ ]]

Out[101]//MatrixForm=

$\begin{pmatrix} 6 & -2 & 2 \\ \{1, 2, 1\} & \{2, 0, 1\} & \{-1, 2, 1\} \end{pmatrix}$

$$y_c(t) = \sum_{i=1}^3 c_i e^{\lambda_i t} v_i$$

$$\begin{aligned} \lambda_1 &= 6 \\ \lambda_2 &= -2 \\ \lambda_3 &= 2 \end{aligned}$$

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$v_3 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

4. Compute the complementary solution of  $y' = Ay$  where A is below

$$4+2i = a+bi \quad \begin{bmatrix} i \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{MatrixForm}[\text{Eigensystem}[A = \begin{pmatrix} 4 & 5 & -12 \\ 4 & 8 & -8 \\ 2 & 5 & -6 \end{pmatrix}]]$$

$$\text{Out[105]//MatrixForm} = \begin{pmatrix} 4+2i & 4-2i & -2 \\ \{1, 2, 1\} & \{-1, 2, 1\} & \{2, 0, 1\} \end{pmatrix}$$

$$y_c(t) = \sum_{i=1}^3 c_i y_i(t)$$

$$y_3 = e^{-2t} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \quad p = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \quad q = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$y_1 = e^{at} (\cos(bt)p - \sin(bt)q)$$

$$y_2 = e^{at} (\sin(bt)p + \cos(bt)q)$$

$y_1$  and  $y_2$  get big and wiggle

5. Compute the complementary solution of  $y' = Ay$  where A is below

In[154]:=

$$\text{MatrixForm}[\text{Eigensystem}[A = \begin{pmatrix} -6 & -5 & 8 \\ 4 & -2 & -8 \\ 2 & 0 & -6 \end{pmatrix}]]$$

$$\text{Out[154]//MatrixForm} = \begin{pmatrix} -6+2i & -6-2i & -2 \\ \{1, 2, 1\} & \{-1, 2, 1\} & \{2, 0, 1\} \end{pmatrix}$$

$$y_c(t) = c_1 y_1 + c_2 y_2 + c_3 y_3$$

$$y_3 = e^{-2t} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$y_1 = e^{-6t} (\cos(2t)p - \sin(2t)q)$$

$$y_2 = e^{-6t} (\sin(2t)p + \cos(2t)q)$$

$$p = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} \quad q = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$y_3$  most imp  $t \rightarrow \infty$

solutions decay to zero like  $e^{-2t}$

6. Compute the complementary solution of  $y' = Ay$  where A is below

In[155]:=

$$\text{MatrixForm}[\text{Eigensystem}[A = \begin{pmatrix} 5 & -1 \\ 4 & 1 \end{pmatrix}]]$$

$$\text{Out[155]//MatrixForm} = \begin{pmatrix} 3 & 3 \\ \{1, 2\} & \{0, 0\} \end{pmatrix} \quad \lambda_1 = 3$$

$$y_c(t) = c_1 e^{3t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 y_2$$

$$y_1 = e^{3t} v_1 \quad v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$y_2 = t e^{3t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + e^{3t} u$$

$$[A - \lambda_1 I] u = v_1$$

$$\begin{bmatrix} 5-3 & -1 \\ 4 & 1-3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

7. Compute the complementary solution of  $y' = Ay$  where A is below

In[165]:=

$$\text{MatrixForm}[\text{Eigensystem}[A = \begin{pmatrix} 5 & -1 \\ 1 & 3 \end{pmatrix}]]$$

$$\text{Out[165]//MatrixForm} = \begin{pmatrix} 4 & 4 \\ \{1, 1\} & \{0, 0\} \end{pmatrix}$$

$$y_c(t) = c_1 e^{4t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 y_2$$

$$y_2 = t e^{4t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{4t} u$$

$$[A - 4I] u = v$$

$$\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$c_2 y_2$  most important  $\sim t e^{4t} u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

# Undetermined Coefficients

## Section 3.6

1. For the ODE  $y' = Ay + b$  with  $b = e^{3t} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$

1.1. Write down the form for the particular solution

$$y_p(t) = ue^{3t}$$

1.2. Write down the Lin System for the coefficients

$$\left[ 3I - A \mid \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right]$$

2. For the ODE  $y' = Ay + b$  with  $b = \cos(4t) \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \sin(4t) \begin{pmatrix} 4 \\ 2 \end{pmatrix}$

2.1. Write down the form for the particular solution

$$y_p(t) = \cos(4t)u_1 + \sin(4t)u_2$$

2.2. Write down the Lin System for the coefficients

$$\left[ \begin{array}{cc|c} -4I & -A & \begin{pmatrix} 4 \\ 2 \end{pmatrix} \\ -A & 4I & \begin{pmatrix} 1 \\ 3 \end{pmatrix} \end{array} \right]$$

3. For the ODE  $y' = Ay + b$  with  $b = \begin{pmatrix} t^2 \\ 1 \\ 2 \end{pmatrix}$

3.1. Write down the form for the particular solution

$$y_p(t) = u_0 + tu_1 + t^2u_2$$

3.2. Write down the Lin System for the coefficients

$$\left[ \begin{array}{ccc|c} -A & I & 0 & \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \\ 0 & -A & 2I & \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ 0 & 0 & -A & \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \end{array} \right]$$

$$\begin{aligned} y_p' &= 3ue^{3t} \\ 3ue^{3t} &= A(ue^{3t}) + e^{3t} \begin{pmatrix} 1 \\ 3 \end{pmatrix} \\ 3u &= Au + \begin{pmatrix} 1 \\ 3 \end{pmatrix} \\ (3I - A)u &= \begin{pmatrix} 1 \\ 3 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} y_p' &= -4\sin(4t)u_1 + 4\cos(4t)u_2 \\ &= \cos(4t)Au_1 + \sin(4t)Au_2 + \cos(4t) \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \sin(4t) \begin{pmatrix} 4 \\ 2 \end{pmatrix} \\ -4u_1 &= Au_2 + \begin{pmatrix} 4 \\ 2 \end{pmatrix} \\ 4u_2 &= Au_1 + \begin{pmatrix} 1 \\ 3 \end{pmatrix} \\ -4Iu_1 - Au_2 &= \begin{pmatrix} 4 \\ 2 \end{pmatrix} \\ -Au_1 + 4Iu_2 &= \begin{pmatrix} 1 \\ 3 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} y_p' &= u_1 + 2tu_2 \\ &= \begin{pmatrix} Au_0 \\ 0 \\ 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}t + t^2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + t^2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} u_1 &= Au_0 + \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \\ 2u_2 &= Au_1 + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ 0 &= Au_2 + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \end{aligned}$$

4. For the ODE  $y' = Ay + b$  with  $b = \begin{pmatrix} e^{2t} \\ -e^{2t} \\ 3e^{2t} \end{pmatrix}$

4.1. Write down the form for the particular solution

$$y_p(t) = ue^{2t}$$

4.2. Write down the Lin System for the coefficients

$$2u = Au + \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$$

5. For the ODE  $y' = Ay + b$  with  $b = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$

5.1. Write down the form for the particular solution

$$y_p(t) = u$$

5.2. Write down the Lin System for the coefficients

$$Au = -b.$$

$$\begin{aligned} y_p' &= 2ue^{2t} \\ &= e^{2t}Au + e^{2t}\begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \\ 2u &= Au + \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \\ (2I - A)u &= \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \end{aligned}$$

$$Au + b = 0$$

$$\begin{aligned} Au &= -b \\ u &= -A^{-1}b \end{aligned}$$

Lin  
#8

$$ty' + 2y = 2t^2$$

$$y(1) = 2$$

$$y' + \frac{2}{t}y = 2t$$

$$v = e^{\int \frac{2}{t} dt} = t^2$$

$$t^2 y' + 2ty = 2t^3$$

$$(t^2 y)' = 2t^3$$

$$t^2 y = \int 2t^3 dt + C$$

$$t^2 y = \frac{2t^4}{4} + C$$

$$t^2 y =$$

$$t^2(2) = \frac{2 \cdot 1^4}{4} + C$$

$$C = 2 - \frac{1}{2} = \frac{3}{2}$$

$$y = \frac{2t^4}{4t^2} + \frac{C}{t^2} = \frac{1}{2}t^2 + \frac{3}{2}t^{-2}$$

Lin

#10

$$t y' + 3y = 3t^3$$

$$\boxed{y(1) = -1}$$

$$y' + \frac{3}{t} y = 3t^2$$

std Form

$$v = e^{\int \frac{3}{t} dt} = e^{3 \ln(t)}$$

$$v = e^{\ln(t^3)} = t^3$$

$$t^3 y' + 3t^2 y = 3t^5$$

$$(t^3 y)' = 3t^5$$

$$t^3 y = \int 3t^5 dt + C$$

$$t^3 y = \frac{3t^6}{6} + C$$

$$y = \frac{3t^6}{6t^3} + Ct^{-3}$$

$$y(1) = -1$$

$$-1 = \frac{3}{6} + C$$

$$\boxed{C = -1 - \frac{1}{2}} = -\frac{3}{2}$$

$$y = \frac{3t^6}{6t^3} - \frac{3}{2} t^{-3}$$